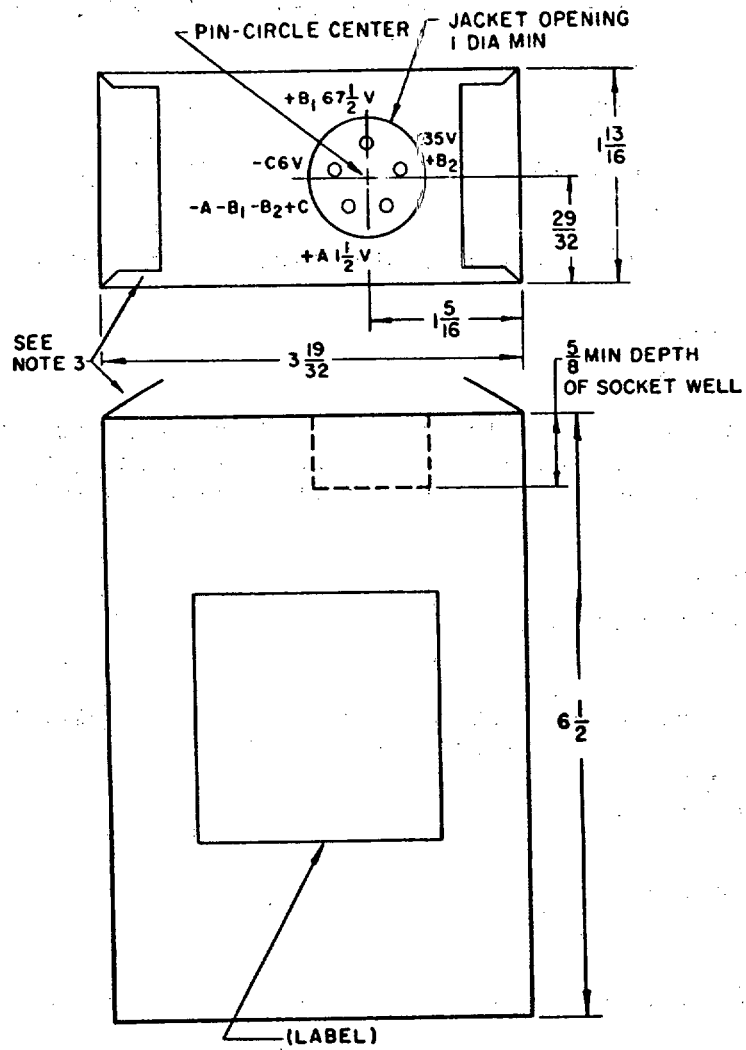
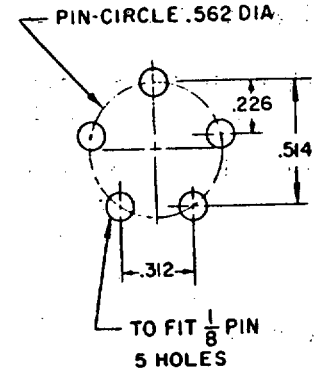
MILITARY SPECIFICATION SHEET

BATTERY, DRY, BA-221/U

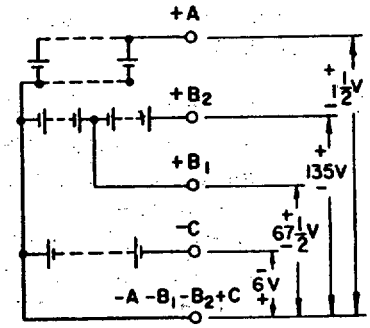
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the dry battery described herein shall consist of this document and the latest issue of Specification MIL-B-18.





SOCKET DETAIL



TYPICAL SCHEMATIC

INCHES	MM	INCHES	MM
.005	.13	5/8	15.88
1/16	1.59	29/32	23.02
1/8	3.18	1	25.40
.226	5.74	1 5/16	33.34
.312	7.92	1 13/16	46.04
.514	13.06	3 19/32	91.28
.562	14.27	6 1/2	165.10

Ⓓ denotes changes

FSC-6135

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:45Z
Check the source to verify that this is the current version before use.

NOTES:

1. All dimensions shown on figure are in inches. Unless otherwise specified, tolerances are ±1/16 on fractions and ±0.005 on decimals.

2. Metric equivalents specified in table are for general information only and are based upon 1 inch = 25.4 millimeters.

3. The two flaps on top of the battery shall be partially glued or taped to the battery jacket.

REQUIREMENTS:

Dimensions and configuration: See figure.

	A unit	B_1 unit	B_2 unit	C unit
Nominal voltage:	1.5 volts	67.5 volts	135 volts	-6 volts
Usual number and type of cells:	8"M"	---	90 "N"	4 "M"
Usual cell connections	Parallel	---	Series 1/	Series

1/ B_2 unit is center tapped to B_1 unit.

Terminals: 5-hole socket.

Weight (Maximum) 2 pounds, 4 ounces.

(D) Capacity tests: When the battery is tested in accordance with the methods of examination and test of this Specification, the minimum capacity-test requirements shall be not less than the minimum time specified for SLD or SLT.

(D) First Article inspection:

Visual and mechanical (External).
Battery voltage.
Vibration test.
Mechanical-shock test per MIL-STD-202, method 213, test condition I.
Insulation resistance test per MIL-STD-202, method 302, test condition Letter B, tolerance ±20 volts.
Capacity, D (without storage).
Jacket integrity test.

(D) Quality conformance inspection:

Visual and mechanical (External).
Battery voltage.
Insulation-resistance test per MIL-STD-202, method 302, test condition Letter B, tolerance ±20 volts.
Vibration test.
Mechanical-shock test per MIL-STD-202, method 213, test condition I.
Jacket integrity test.
Electrode leakage.
Capacity, D & T

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:45Z
Check the source to verify that this is the current version before use.

Methods of examination and test:

Capacity tests: See requirements for capacity specified herein.

Storage -

Test	Period
D	6 months
T	90 days

Discharge - All units shall be discharged concurrently through the resistances listed below for 2 minutes on each load, alternating on loads 1 and 2. This cycle shall be repeated continuously to the applicable test-end voltage listed below.

Resistance	A unit	B_1 unit	B_2 unit	C unit
Load 1	6.5 ohms	5,200 ohms	Open circuit	Open circuit
Load 2	6.5 ohms	16,900 ohms	5,870 ohms	200 ohms
Test-end voltage:	1.0 volts	45.0 volts	90.0 volts	-4.0 volts

Closed circuit voltage - TS-183( )/U test per special marking or, with minimum permissible voltage specified for TS-183( )/U, use load resistance values noted below:

A unit	5.983 ohms
B_1 unit	DO NOT TEST
B_2 unit	4,879 ohms
C unit	298.2 ohms

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:45Z
Check the source to verify that this is the current version before use.

Special marking on each unit package:

TO TEST THIS BATTERY WITH TS-183 ( )/U:

UNIT	USE JACK NO.	MINIMUM PERMISSIBLE VOLTAGE
A	3	1.38
B_2	27	124.2
C	12	-5.52

Unless expressly authorized. this test information shall apply to Army and Air Force applications only.

Custodians:

Army - ER
Navy - SH
Air Force - 99

Review Activities:

Army - MI, AR

User Activities:

Navy - YD, MC

Preparing activity:

Army - ER

Project number: (6135-0155-32)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:45Z
Check the source to verify that this is the current version before use.